

Tae Ha "Jeff" Park

GN&C RESEARCH ENGINEER, NARA SPACE TECHNOLOGY

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Education

Stanford University

Ph.D. in Aeronautics & Astronautics | Advisor: Dr. Simone D'Amico

Stanford, CA
04/2018 – 06/2024

- Thesis: Robust Machine Learning for Vision-Based Navigation about Non-Cooperative Resident Space Objects
- Teaching Assistant for AA279A: Space Mechanics (2019, 2021, 2022).

Stanford University

M.S. in Aeronautics & Astronautics

Stanford, CA
09/2017 – 04/2020

- Conducted research on deep learning, computer vision, spacecraft swarm navigation and optimization

Harvey Mudd College (HMC)

B.S. in Engineering

Claremont, CA
08/2013 – 05/2017

- Graduated with High Distinction (GPA: 3.81/4.0)
- Member of the Tau Beta Pi Engineering Honors Society
- De Pietro fellow in Civil Engineering

Experience

Nara Space Technology Inc.

Project Manager | Development of AI-Based Space Traffic Management Technology using Microsatellites

Seoul, Korea
12/2024 – Current

- PM of a 3-year R&D project (₩1,200,000,000) awarded by the Ministry of SMEs and Startups of South Korea on the topic of AI-empowered vision-based rendezvous and proximity operations with unknown non-cooperative spaceborne targets

Research Engineer | Guidance, Navigation and Control (GN&C) Team I

07/2024 – Current

- Developing an in-house CubeSat AOCS system and its software & hardware validation environments
- Developing an intelligent system for autonomous vision-based navigation about arbitrary resident space objects

Infinite Orbits SAS

Computer Vision and GN&C Intern

Toulouse, France
06/2022 – 08/2022

- Constructed a satellite rendezvous simulator and scene renderer based on Unreal Engine 5 and C++ to train and validate convolutional neural networks for monocular pose estimation and tracking of known noncooperative spacecraft

Space Rendezvous Laboratory (SLAB)

Research Assistant | Advisor: Dr. Simone D'Amico

Stanford, CA
01/2019 – 06/2024

- Developed the Testbed for Rendezvous and Optical Navigation (TRON) facility that can simulate various Rendezvous and Proximity Operation (RPO) scenarios using a mockup satellite model under high-fidelity spaceborne illumination settings [C3]
- Developed advanced open-source benchmark datasets (e.g., SPEED+, SHIRT, SPE3R) using synthetic renders—OpenGL/Unreal Engine—and TRON to train and validate spaceborne vision-based Deep Learning (DL) and navigation algorithms with emphasis on robustness across domain gap between synthetic and spaceborne data [C1, C4, C6, C8, C9, D2–D5, J2, J3, J5]
- Developed and integrated robust DL models and navigation algorithms for vision-based RPO with noncooperative target [C5–C7, C10, J4–J6]
- Developed a DL model for single-shot 3D reconstruction and pose estimation of an unknown spacecraft from 2D images [C8]

Publications

🔗 Link to arXiv/official paper/project webpage | 📁 GitHub repository | 📄 Link to paper PDF | 📺 Youtube video

Preprints

[P1] Bridging Domain Gap for Flight-Ready Spaceborne Vision

T. H. Park, S. D'Amico

arXiv preprint arXiv:2409.11661 (2024). 🔗

Peer-Reviewed Journal Articles

- [J6] Robust Multi-Task Learning and Online Refinement for Spacecraft Pose Estimation across Domain Gap
T. H. Park, S. D'Amico
Advances in Space Research (2024). [🔗](#) [🔄](#)
- [J5] Adaptive Neural-Network-Based Unscented Kalman Filter for Robust Pose Tracking of Noncooperative Spacecraft
T. H. Park, S. D'Amico
Journal of Guidance, Control, and Dynamics (2023). [🔗](#)
- [J4] Leveraging Neural Network Uncertainty in Adaptive Unscented Kalman Filter for Spacecraft Pose Estimation
L. Pasqualetto Cassinis, **T. H. Park**, N. Stacey et al.
Advances in Space Research (2023). [🔗](#)
- [J3] Satellite Pose Estimation Competition 2021: Results and Analyses
T. H. Park, M. Märten, M. Jawaid et al.
Acta Astronautica (2023). [🔗](#)
- [J2] Satellite Pose Estimation Challenge: Dataset, Competition Design and Results
M. Kisantal, S. Sharma, **T. H. Park** et al.
IEEE Transactions on Aerospace and Electronic Systems (2020). [🔗](#) <**M. BARRY CARLTON AWARD**>
- [J1] Improved Modal Convergence Using the Assumed Modes Method for Rods Carrying Various Lumped Elements
P. D. Cha, **T. H. Park**
International Journal of Mechanical Engineering Education (2018). [🔗](#)

Conference Proceedings & Presentations

- [C10] Online Supervised Training of Spaceborne Vision during Proximity Operations using Adaptive Kalman Filtering
T. H. Park, S. D'Amico
2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan (2024). [🔗](#)
- [C9] SPEED-UE-Cube: A Machine Learning Dataset for Autonomous, Vision-Based Spacecraft Navigation
Z. Ahmed, **T. H. Park**, A. Bhattacharjee et al.
46th AAS Guidance, Navigation and Control Conference, Breckenridge, Colorado, February 2-7 (2024). [🔗](#) [🔄](#)
- [C8] Rapid Abstraction of Spacecraft 3D Structure from Single 2D Image
T. H. Park, S. D'Amico
2024 AIAA SciTech Forum (2024). [🔗](#) [🔗](#) (Correction) [🔄](#)
- [C7] Adaptive End-to-End Architecture for Autonomous Spacecraft Navigation and Control During Rendezvous and Proximity Operations
J. Kruger, T. Guffanti, **T. H. Park** et al.
2024 AIAA SciTech Forum (2024). [🔗](#)
- [C6] Adaptive Neural Network-based Unscented Kalman Filter for Spacecraft Pose Tracking at Rendezvous
T. H. Park, S. D'Amico
2022 AAS/AIAA Astrodynamics Specialist Conference, Charlotte, North Carolina, August 7-11 (2022). [🔗](#)
- [C5] Robust Multi-Task Learning and Online Refinement for Spacecraft Pose Estimation across Domain Gap
T. H. Park, S. D'Amico
11th International Workshop on Satellite Constellations & Formation Flying, Milano, Italy, June 7-10 (2022). [🔗](#)
- [C4] SPEED+: Next-Generation Dataset for Spacecraft Pose Estimation across Domain Gap
T. H. Park, M. Märten, G. Lecuyer et al.
2022 IEEE Aerospace Conference (2022). [🔗](#) [🔄](#) [📺](#)
- [C3] Robotic Testbed for Rendezvous and Optical Navigation: Multi-Source Calibration and Machine Learning Use Cases
T. H. Park, J. Bosse, S. D'Amico
2021 AAS/AIAA Astrodynamics Specialist Conference, Virtual, August 8 - 12 (2021). [🔗](#)
- [C2] Generative Model for Spacecraft Image Synthesis using Limited Dataset
T. H. Park, S. D'Amico

[C1] Towards Robust Learning-Based Pose Estimation of Noncooperative Spacecraft

T. H. Park, S. Sharma, S. D'Amico

2019 AAS/AIAA Astrodynamics Specialist Conference, Portland, Maine, August 11 - 15 (2019).  <BEST PAPER AWARD>

Open-Source Datasets

[D5] Spacecraft Pose Estimation Dataset of a 3U CubeSat using Unreal Engine (SPEED-UE-Cube)

T. H. Park, Z. Ahmed, A. Bhattacharjee et al.

Stanford Digital Repository (2024). 

[D4] SPE3R: Synthetic Dataset for Satellite Pose Estimation and 3D Reconstruction

T. H. Park, S. D'Amico

Stanford Digital Repository (2024).

[D3] SHIRT: Satellite Hardware-In-the-loop Rendezvous Trajectories Dataset

T. H. Park, S. D'Amico

Stanford Digital Repository (2022). 

[D2] Next Generation Spacecraft Pose Estimation Dataset (SPEED+)

T. H. Park, M. Märtens, G. Lecuyer et al.

Stanford Digital Repository (2021).

[D1] Spacecraft Pose Estimation Dataset (SPEED)

S. Sharma, **T. H. Park**, S. D'Amico

Stanford Digital Repository (2019). 

Thesis

[T1] Robust machine learning for vision-based navigation about non-cooperative resident space objects

T. H. Park

Ph.D. Thesis, Stanford University (2024). 

Activities

Co-organizer Kelvins Satellite Pose Estimation Competition (2021) 

AI4Space (CVPR workshop) (2024) 

Reviewer Acta Astronautica (1 paper(s))

Advances in Space Research (7)

Astrodynamics (1)

IEEE Transactions on Aerospace and Electronic Systems (5)

Journal of Aerospace Information Systems (4)

Journal of Guidance, Control, Dynamics (3)

Journal of Spacecraft and Rockets (3)

Skills

Programming MATLAB/Simulink, Python, C/C++, CUDA, $\LaTeX$

Deep Learning PyTorch (Python & C++), ONNXRuntime, TensorRT, MATLAB Deep Learning Toolbox™

Libraries OpenCV, PyTorch3D, OpenMP, CVX/CVXPY

Rendering OpenGL, Unreal Engine

Languages Korean (native), English (fluent), Japanese (proficient), French (conversational)

Honors & Awards

2024	2020 M. Barry Carlton Award , <i>IEEE Aerospace and Electronics System Society</i>	-
2019	Best Paper Award , <i>2019 AAS/AIAA Astrodynamics Specialist Conference</i>	Portland, ME
2015	Tau Beta Pi Engineering Honors Society , <i>HMC</i>	Claremont, CA
2015	De Pietro Fellowship in Civil Engineering , <i>HMC</i>	Claremont, CA
2013	Harvey S. Mudd Merits , <i>HMC</i>	Claremont, CA
2013-17	Dean's List , <i>HMC</i>	Claremont, CA